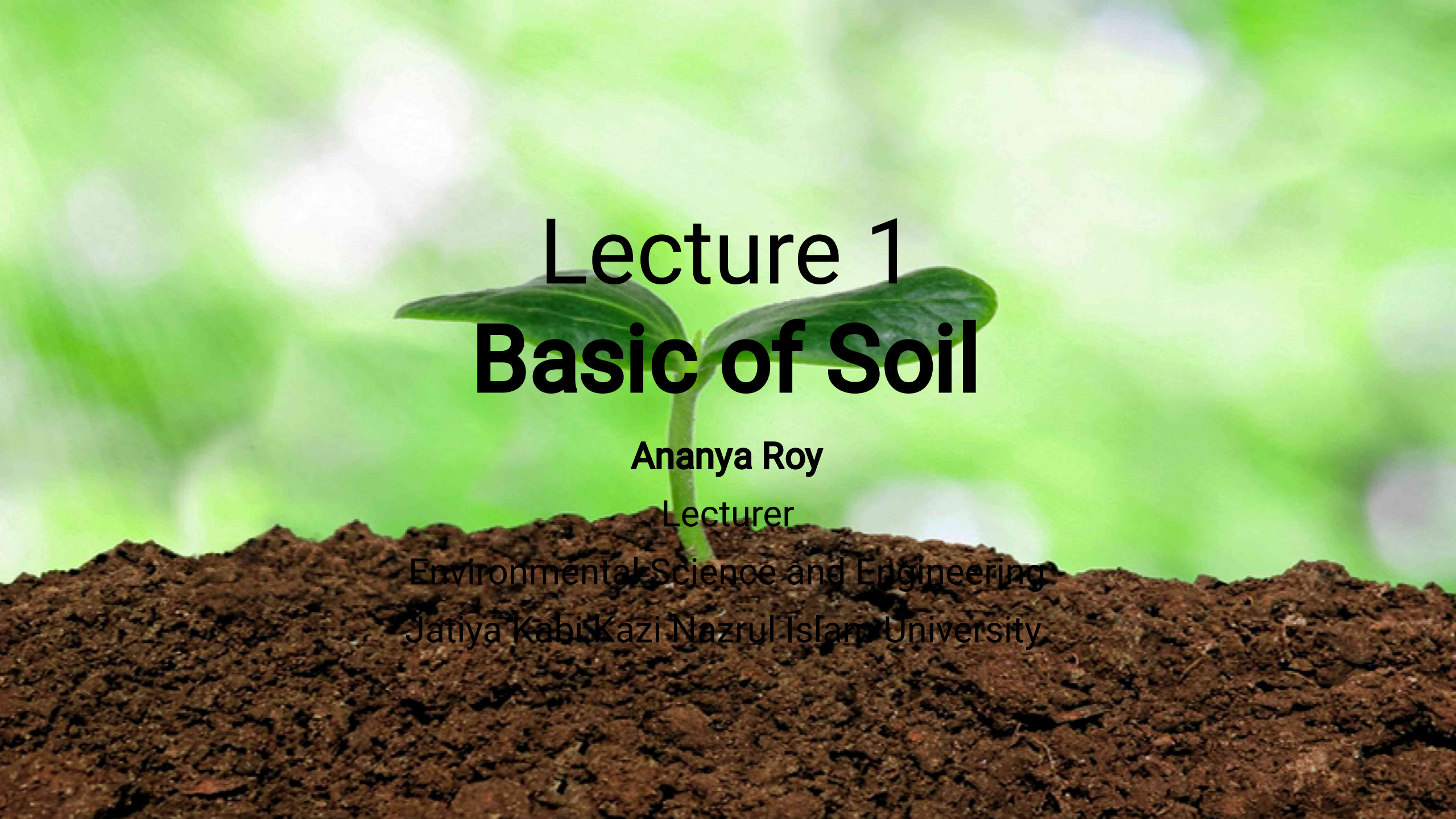




Lecture 1

Basic of Soil

Ananya Roy

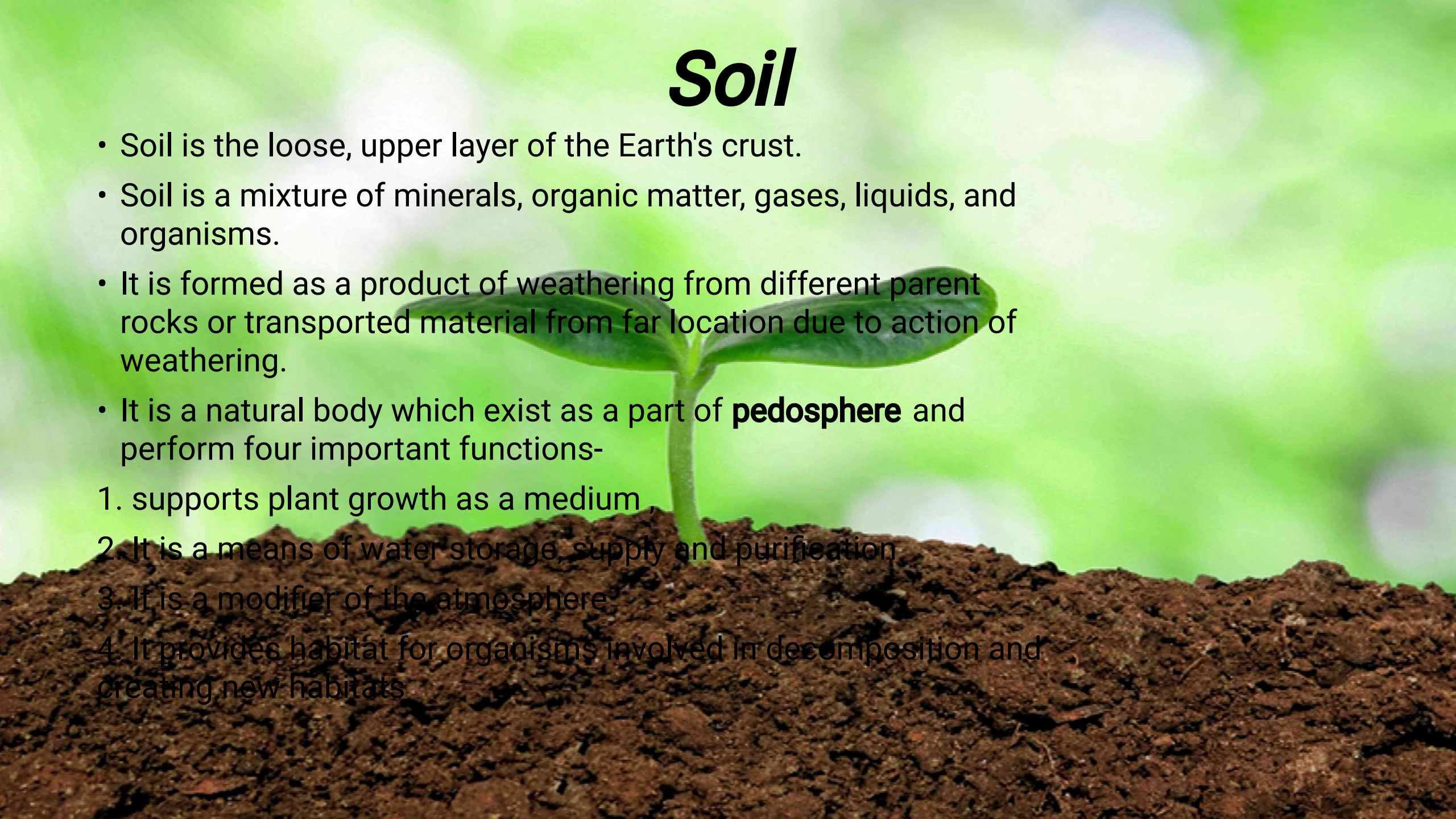
Lecturer

Environmental Science and Engineering

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Soil

- Soil is the loose, upper layer of the Earth's crust.
- Soil is a mixture of minerals, organic matter, gases, liquids, and organisms.
- It is formed as a product of weathering from different parent rocks or transported material from far location due to action of weathering.
- It is a natural body which exist as a part of **pedosphere** and perform four important functions-
 1. supports plant growth as a medium ,
 2. It is a means of water storage, supply and purification
 3. It is a modifier of the atmosphere
 4. It provides habitat for organisms involved in decomposition and creating new habitats



Pedosphere

- The pedosphere is the outermost layer of the Earth that is composed of soil and subject to soil formation processes.
- The study of soil is called pedology.
- The pedosphere **is the skin of the Earth** and only develops when there is a dynamic interaction between the atmosphere (air in and above the soil), biosphere (living organisms), lithosphere (unconsolidated regolith and consolidated bedrock) and the hydrosphere (water in, on and below the soil).

Soil Composition

Mineral Components: Fragments of rock derived from the underlying bedrock and its weathered layers.

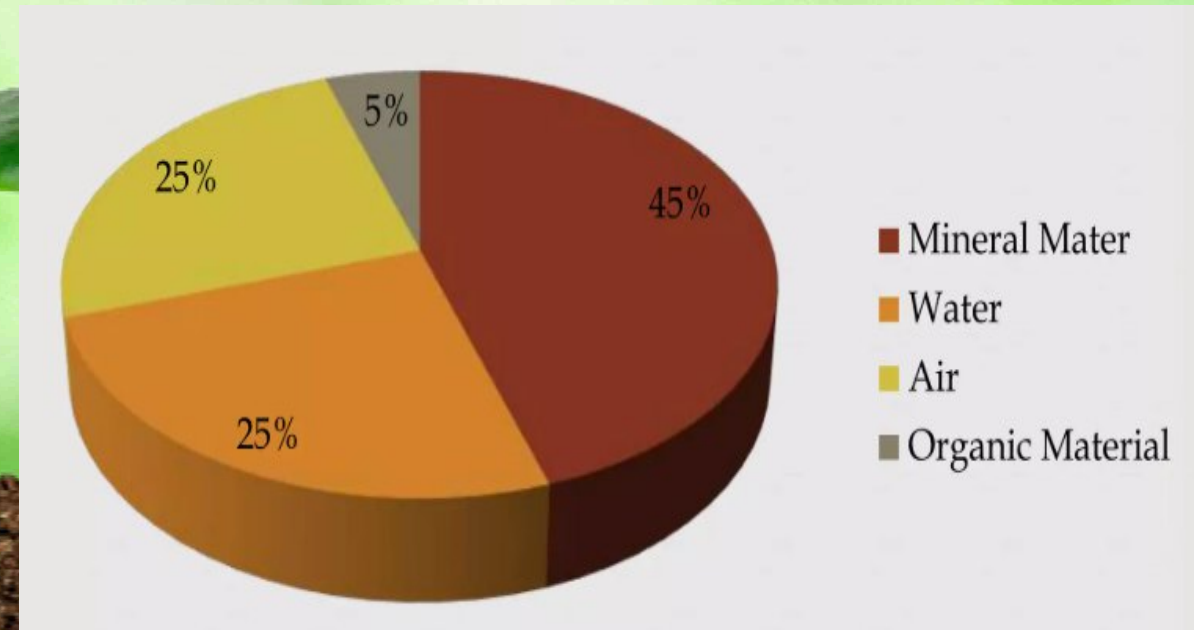
Air: Occupies the tiny spaces between soil particles, allowing for gas exchange.

Water: Also resides within the gaps between soil grains, essential for plant and microbial life.

Organic Matter:

Humus: A dark, sticky substance formed from the decomposition of plant leaves and animal remains.

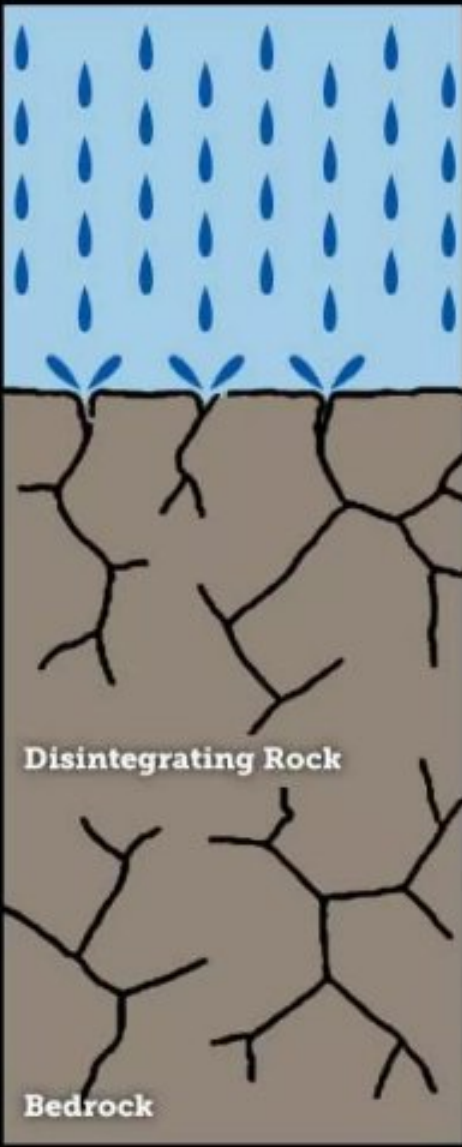
Living Organisms: Includes earthworms, beetles, fungi, and bacteria that contribute to soil health and nutrient cycling.



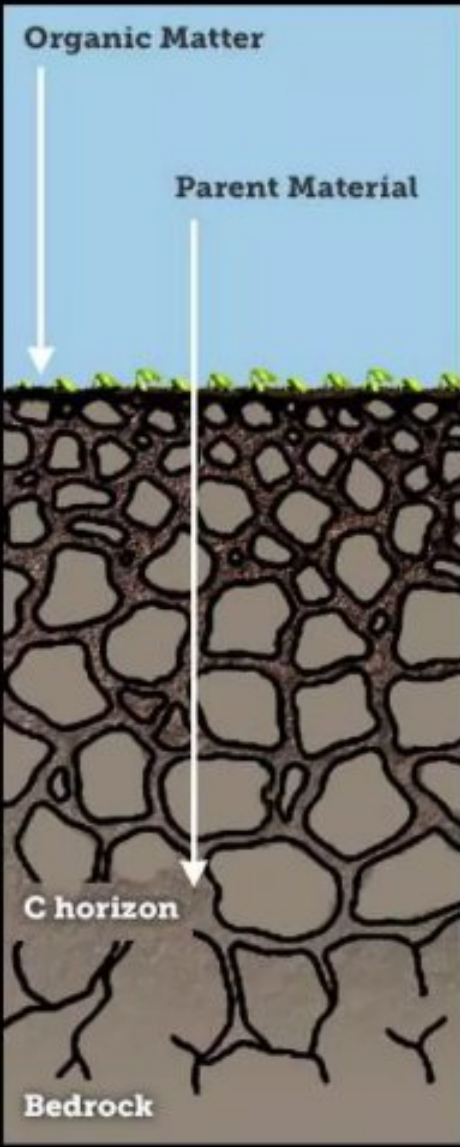
Soil air vs atmospheric air- their composition

Soil Formation

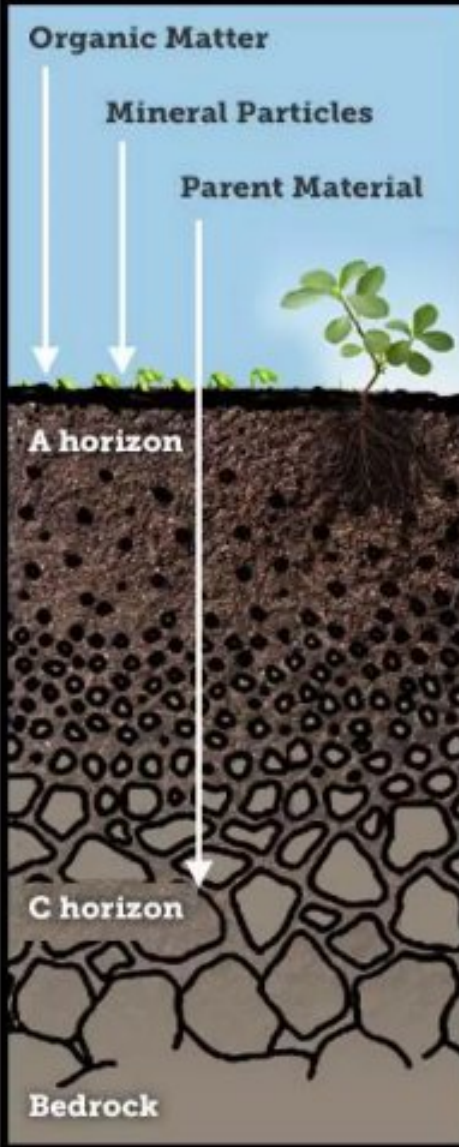
1: Soil Begins to Form



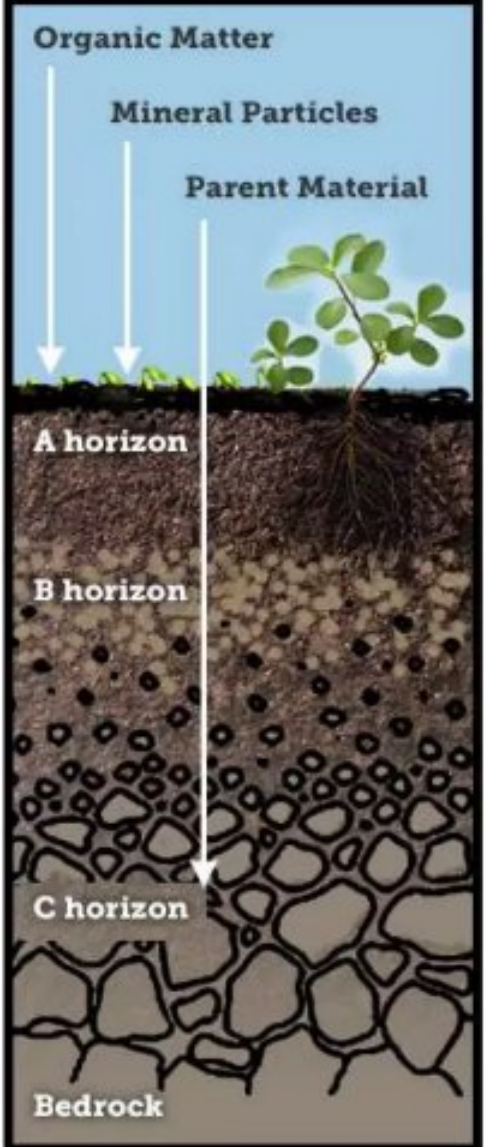
2: Simple Organisms



3: Horizons Form



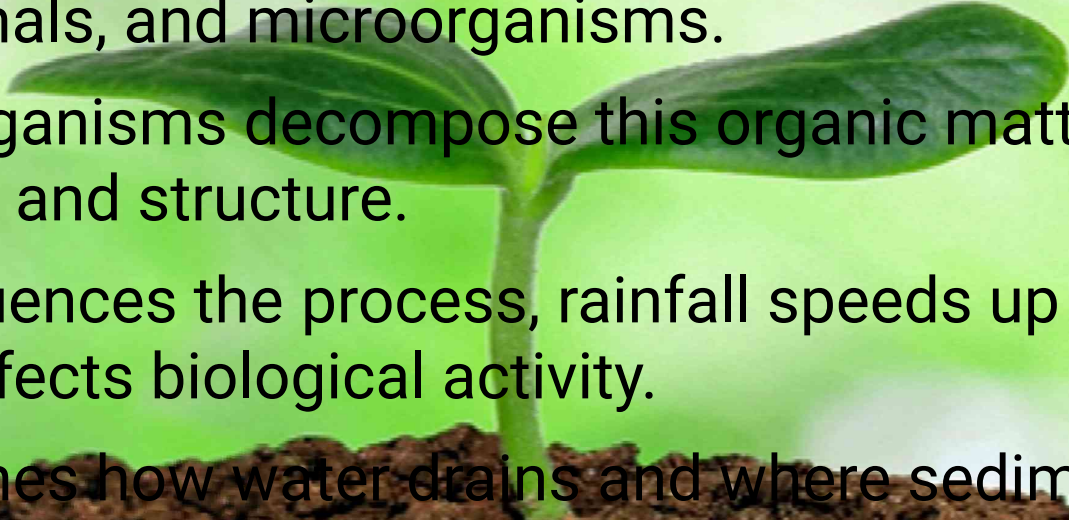
4: Well-developed Soil



Soil formation

- Pedogenesis is the branch where focuses on how soils form and their features.
- It take hundreds of years to develop thin layer of soil
- It can be divided into two major stages-
- **weathering and soil development**
- Weathering process done by three method-**physical, chemical and biological**

Soil formation

- Soil formation starts when parent rocks break down through physical weathering (temperature changes, wind, water), chemical reactions, and biological activity.
 - The tiny rock fragments gradually mix with organic materials produced from the decay of plants, animals, and microorganisms.
 - Microbes and soil organisms decompose this organic matter into humus, which improves soil fertility and structure.
 - Climate strongly influences the process, rainfall speeds up chemical weathering, while temperature affects biological activity.
 - Topography determines how water drains and where sediments accumulate, shaping soil depth and moisture content.
 - Over long periods, continuous weathering and organic accumulation create distinct soil horizons (O, A, B, C).
 - Eventually, these layers form a mature soil profile capable of supporting vegetation and ecological processes.
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Soil Formation

1. Solid rock particles break down from mechanical weathering.

2. Chemical weathering releases important nutrients from the rock grains.

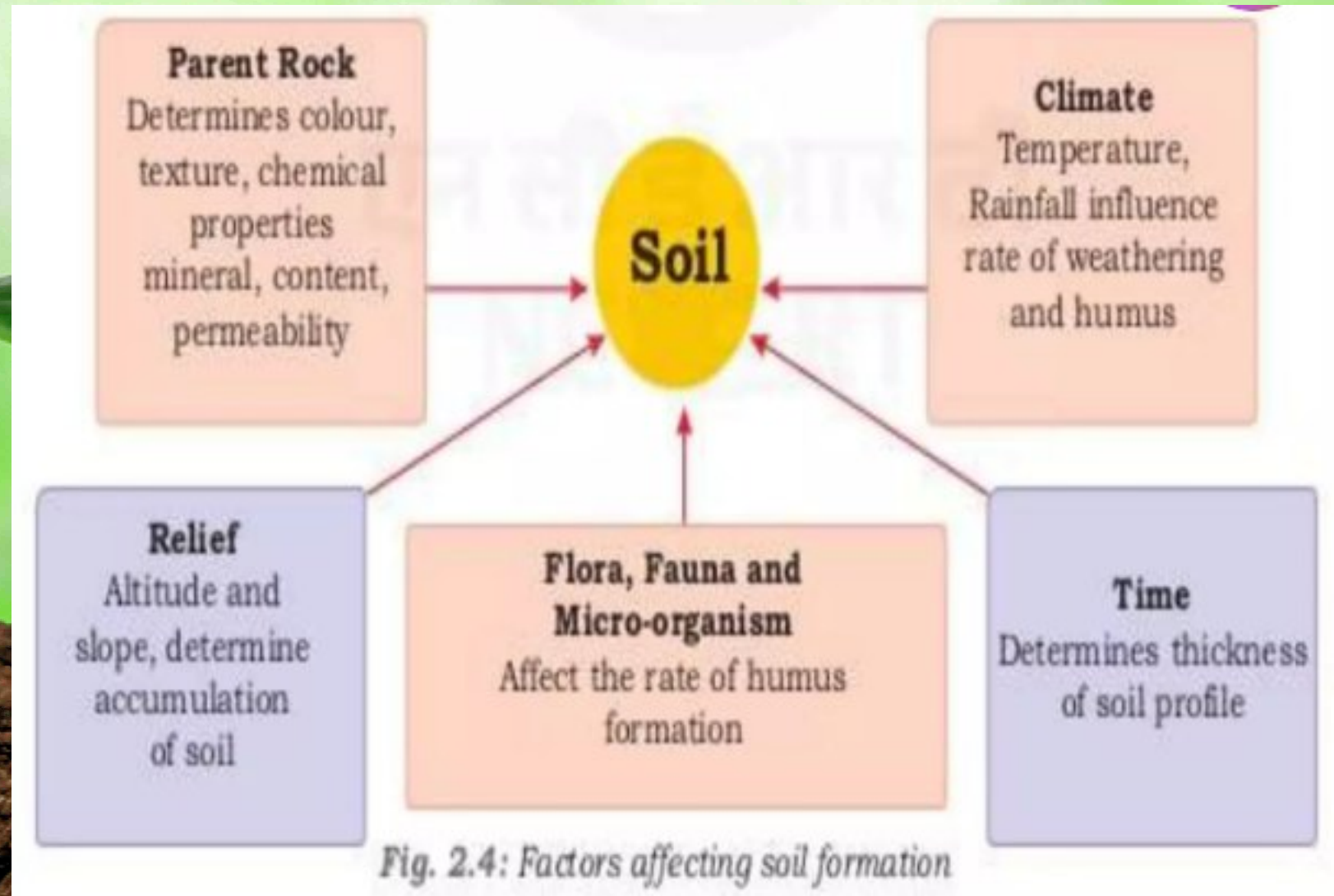
3. Seeds are blown or carried onto the soil grains and may grow into plants that enrich the soil when they die.

4. Micro-organisms decompose the remains of plants to form humus which further enriches the soil.

5. The cycle continues until soils reaches maximum fertility given the climate it is in.

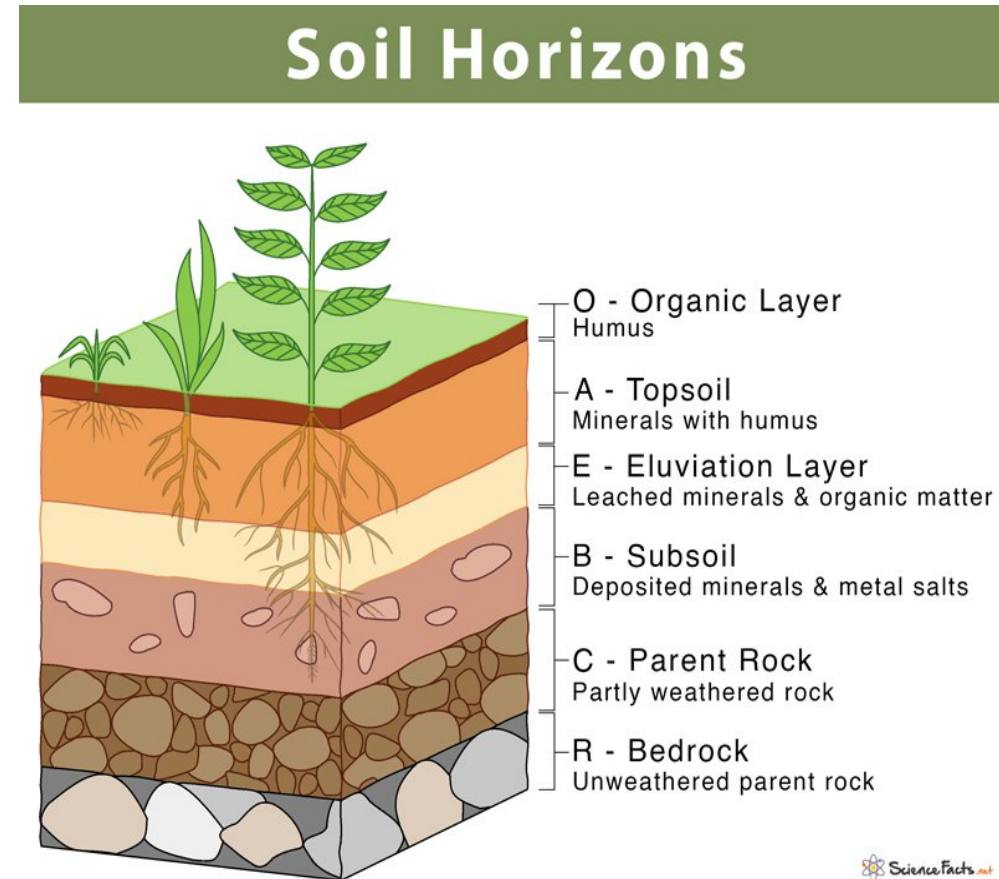
Factors influenced soil formation

1. Parent rock
2. Climate
3. Relief
4. Time
5. Flora, fauna and Micro-organism



Soil Horizon/ Soil Profile

- During its formation, the soil is arranged in different layers.
- Each of these layers is called a soil horizon, and when these layers are arranged sequentially one above the other, it forms the soil profile.
- There are six different layers or horizons that make up a mature soil profile.
- These layers or horizons are represented by alphabets O, A, E, C, B, and R. Immature soils lack some of these layers



Soil Horizon/ Soil Profile

O Horizon – (Organic Layer)

'O' is for organic. This layer is the uppermost layer of the soil rich in organic matter, such as the remains of plants and dead animals. Due to high organic content, this layer is typically black brown or dark brown. The O horizon is thin in some soil, thick in some others, or absent in the rest.

A Horizon – (Topsoil)

Found below the O horizon, it has a dark brown color as it contains the maximum organic matter of the soil. The A horizon or topsoil is thus also called **the humus layer**. The topsoil is the region of intense biological activity and has the most nutrients. Insects, earthworms, centipedes, bacteria, fungi, and other animals are found inside this layer.

The humus makes the topsoil highly porous, allowing it to hold air and moisture necessary for seed germination. Here, the plants stretch their roots deep down, allowing it to hold the topsoil together. In this layer

E Horizon – (Eluviation Layer)

This layer consists of nutrients leached from O and A horizons and is thus called the eluviations layer. Leaching of clay, minerals, and organic matter leaves this layer with a high concentration of sand, silt particles, quartz, and other resistant materials. E horizon is absent in most soils but is more common in forested areas.

B Horizon- (Subsoil Layer)

Mostly found below the topsoil is another layer called the subsoil or horizon B. It is lighter in color than the topsoil due to lower humus content. However, it is comparatively more rigid and compact than the topsoil. This layer has less organic content but is rich in minerals that are leached down from the topsoil. The subsoil is the region of deposition of certain minerals and salts of certain metals such as iron oxides, aluminum oxides, and calcium carbonate in large proportion.

This layer holds enough water due to its clayey nature. Farmers often mix

C Horizon – (Parent Rock)

Also known as **regolith or saprolite**, it lies just below the subsoil. It is called the parent rock because all the upper layers developed from this layer. C horizon is devoid of any organic matter and is made of broken-up bedrocks, making it hard. **Plant roots do not penetrate this layer**. This layer is a transition between the inner layer of earth and the upper A and B horizons.

R Horizon – (Bedrock)

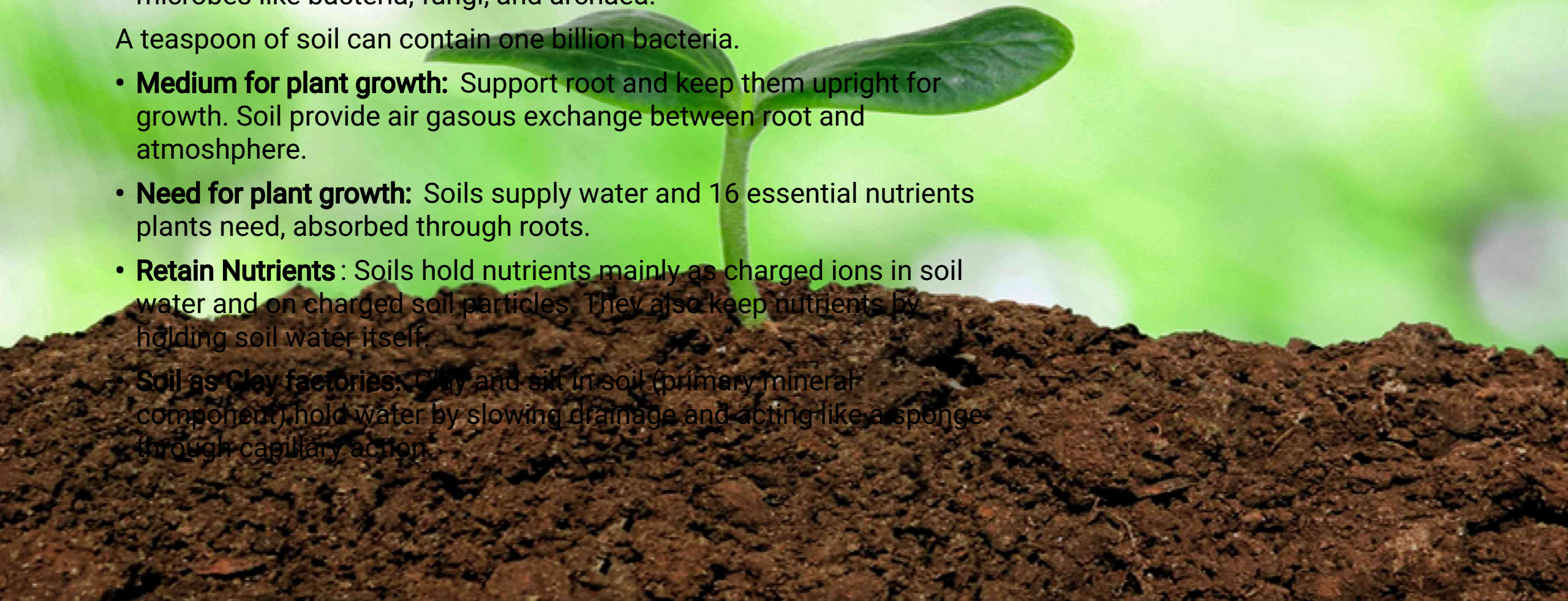
Found beneath all the layers, it consists of un-weathered igneous, sedimentary, and metamorphic rocks. It is highly compact. Granite, basalt, quartzite, sandstone, and limestone make up the bedrock.

Importance of Soil

- **Soils are ecosystems:** They hosting many organisms, including mammals, birds, insects, and protozoa. The richest diversity is in soil microbes like bacteria, fungi, and archaea.

A teaspoon of soil can contain one billion bacteria.

- **Medium for plant growth:** Support root and keep them upright for growth. Soil provide air gasous exchange between root and atmoshphere.
- **Need for plant growth:** Soils supply water and 16 essential nutrients plants need, absorbed through roots.
- **Retain Nutrients :** Soils hold nutrients mainly as charged ions in soil water and on charged soil particles. They also keep nutrients by holding soil water itself.
- **Soil as Clay factories:** Clay and silt in soil (primary mineral component) hold water by slowing drainage and acting like a sponge through capillary action.



Importance of Soil

- **Habitat for insect and other organism:** Soils house many insects and tiny microbes that need soil for food and air. Worms, termites, and others live there, where moisture and air help break down organic matter.
- **Filtration System for surface water:** After rain or snow melts, water flows on the surface but much soaks into the ground. As water moves down through soil layers, it gets filtered, removing dust, chemicals, and pollutants. This makes aquifers one of the cleanest water sources.
- **Store CO₂:** Soils store carbon and help control atmospheric CO₂. During humification, some organic matter resists breakdown, especially in acidic, water-rich soils like peat. Soils hold twice as much carbon as the atmosphere and three times more than plants. This leads to organic matter rich in carbon building up in soil. Soil also stores and cycles nutrients like nitrogen and phosphorus.